Deep Learning Recommendation System cross-GPU Communication Optimization

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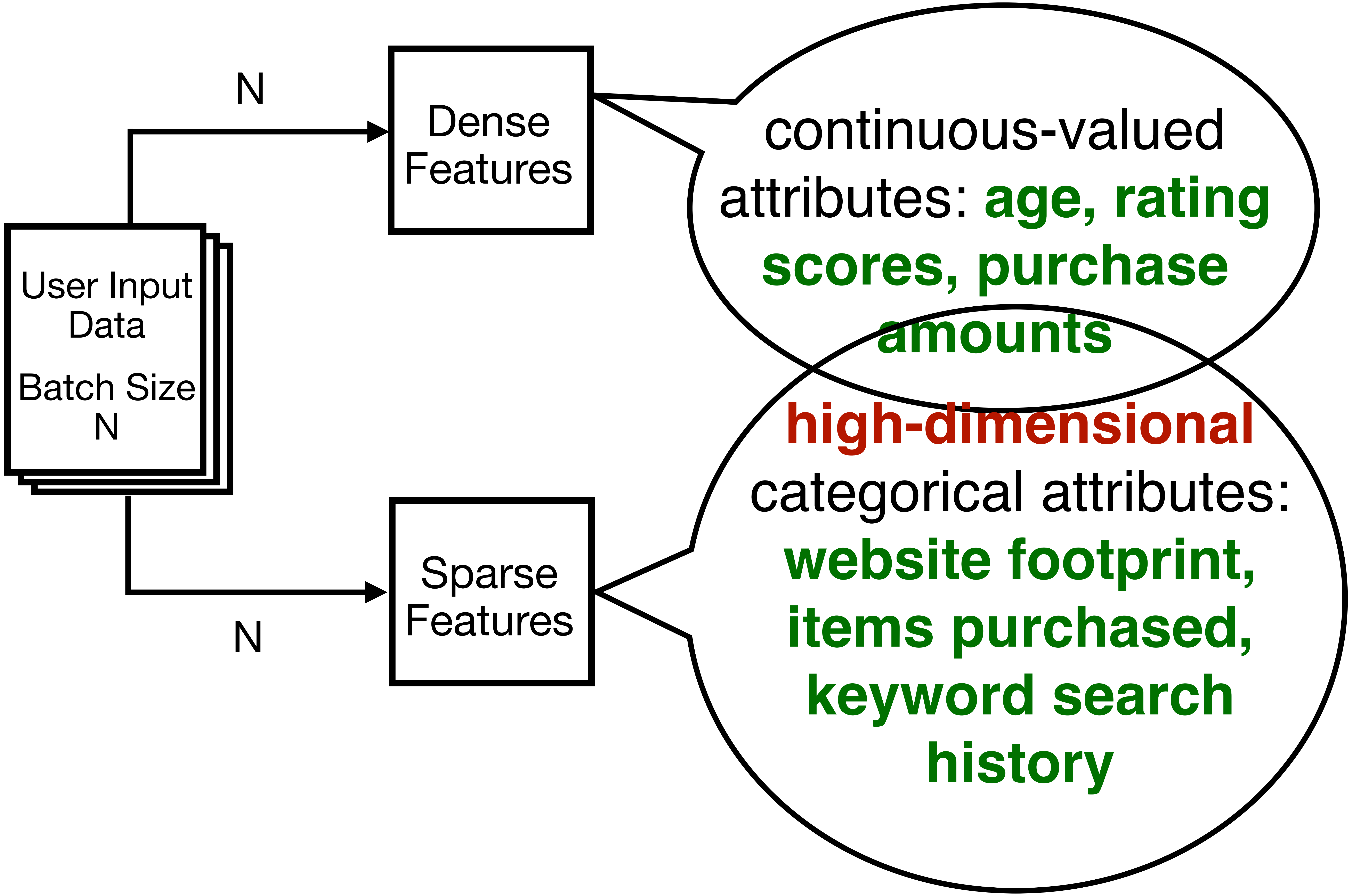
Agenda

1. **Deep Learning Recommendation Model (DLRM)**
2. **DLRM Communication Pattern**
3. **PGAS Async One-sided Small Messages**
4. **Case Studies**

Deep Learning Recommendation Model (DLRM) Inferencing



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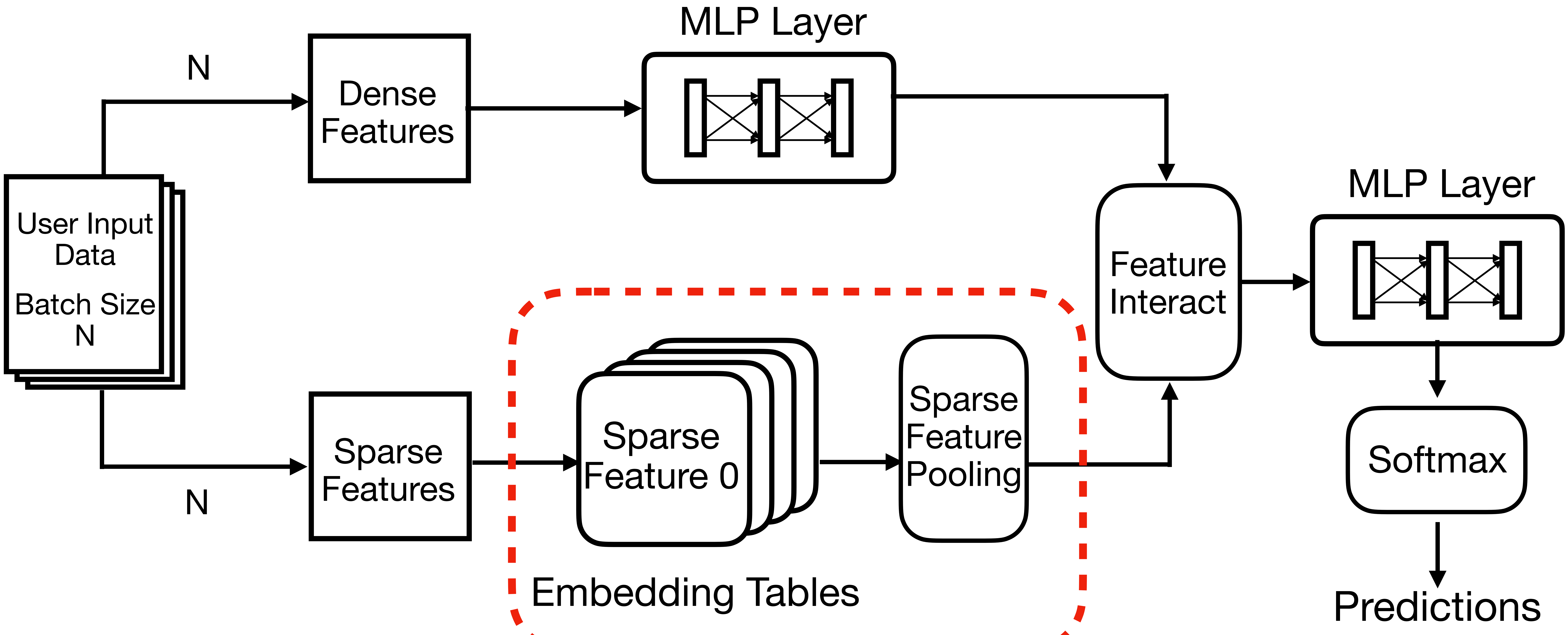

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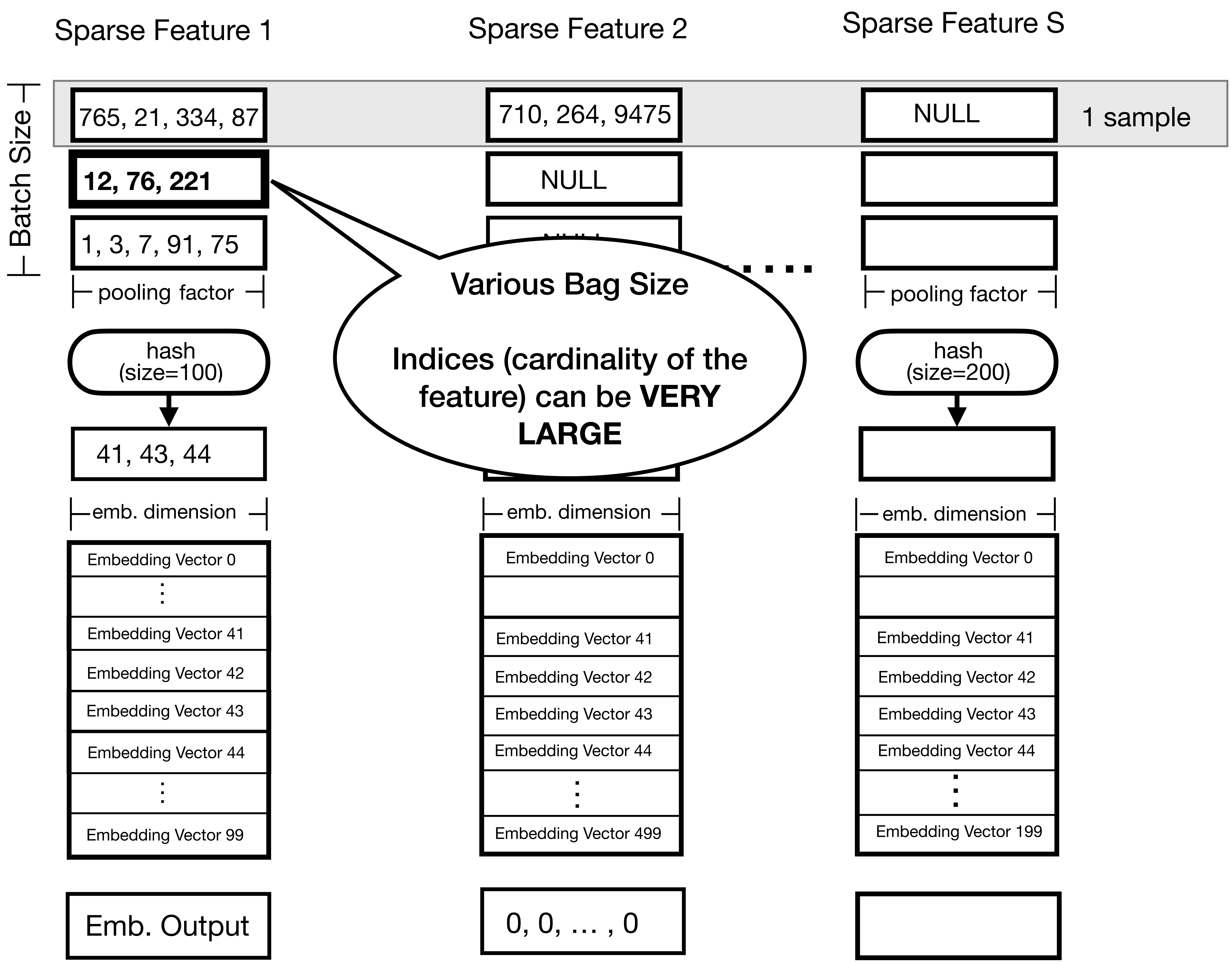

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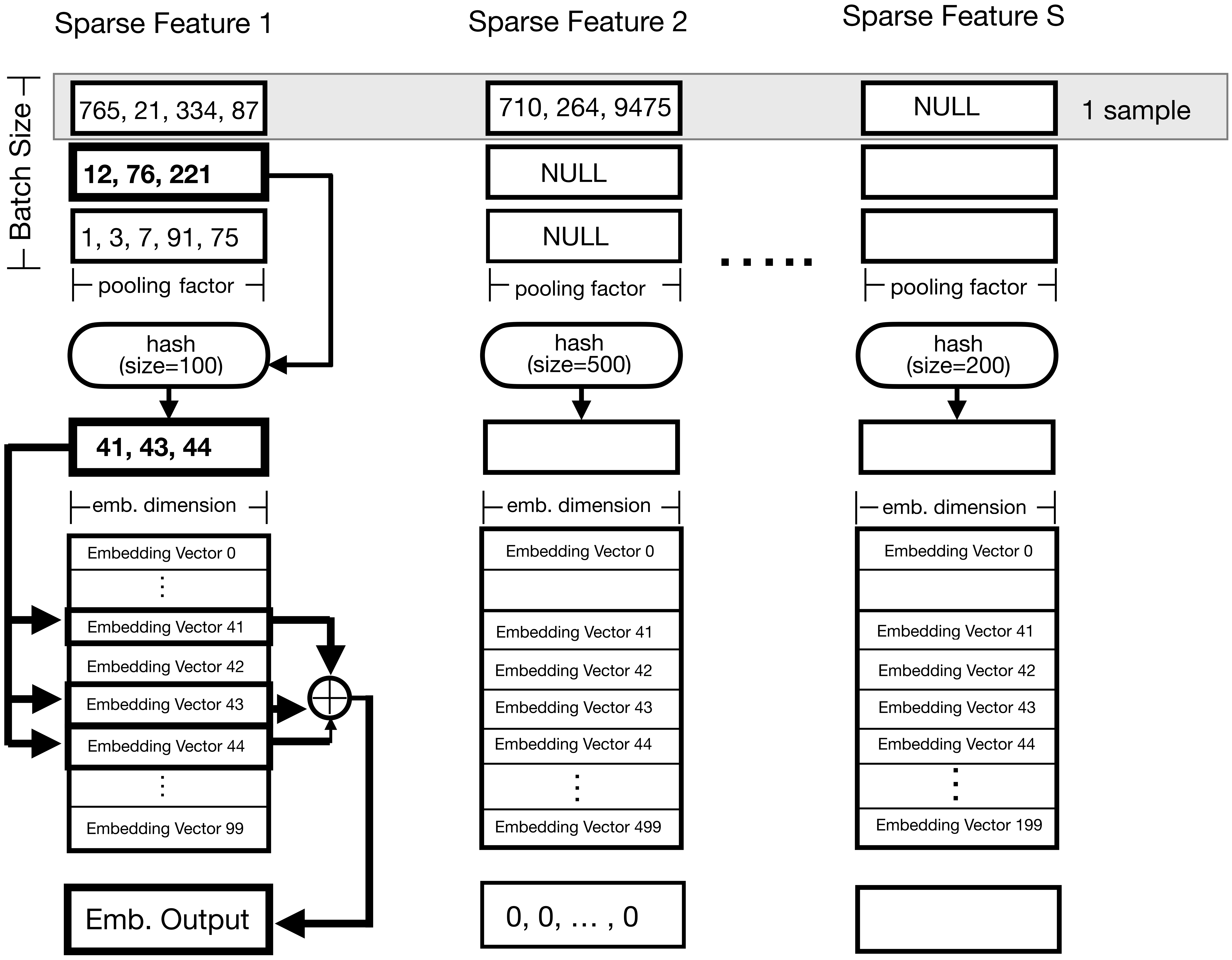

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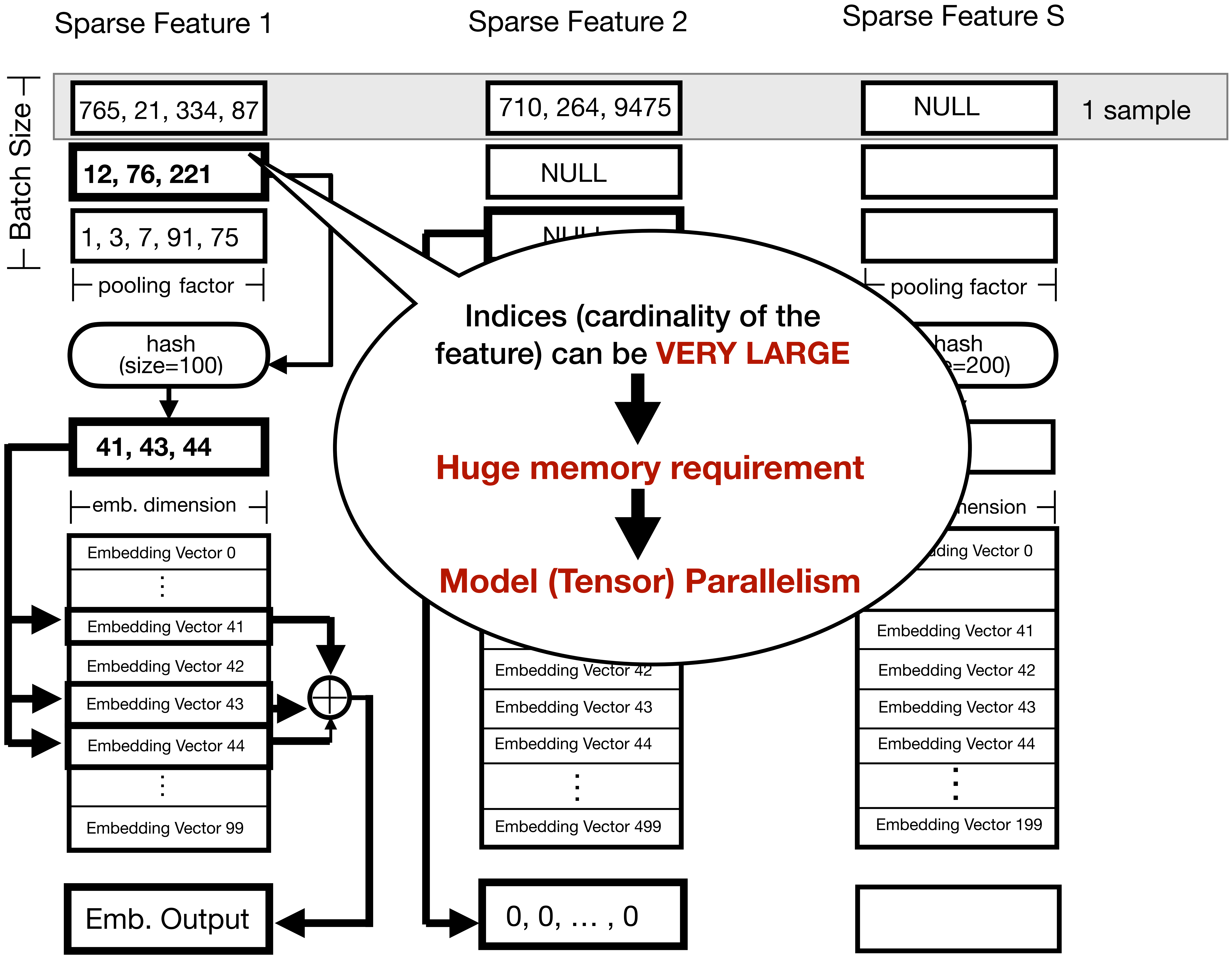

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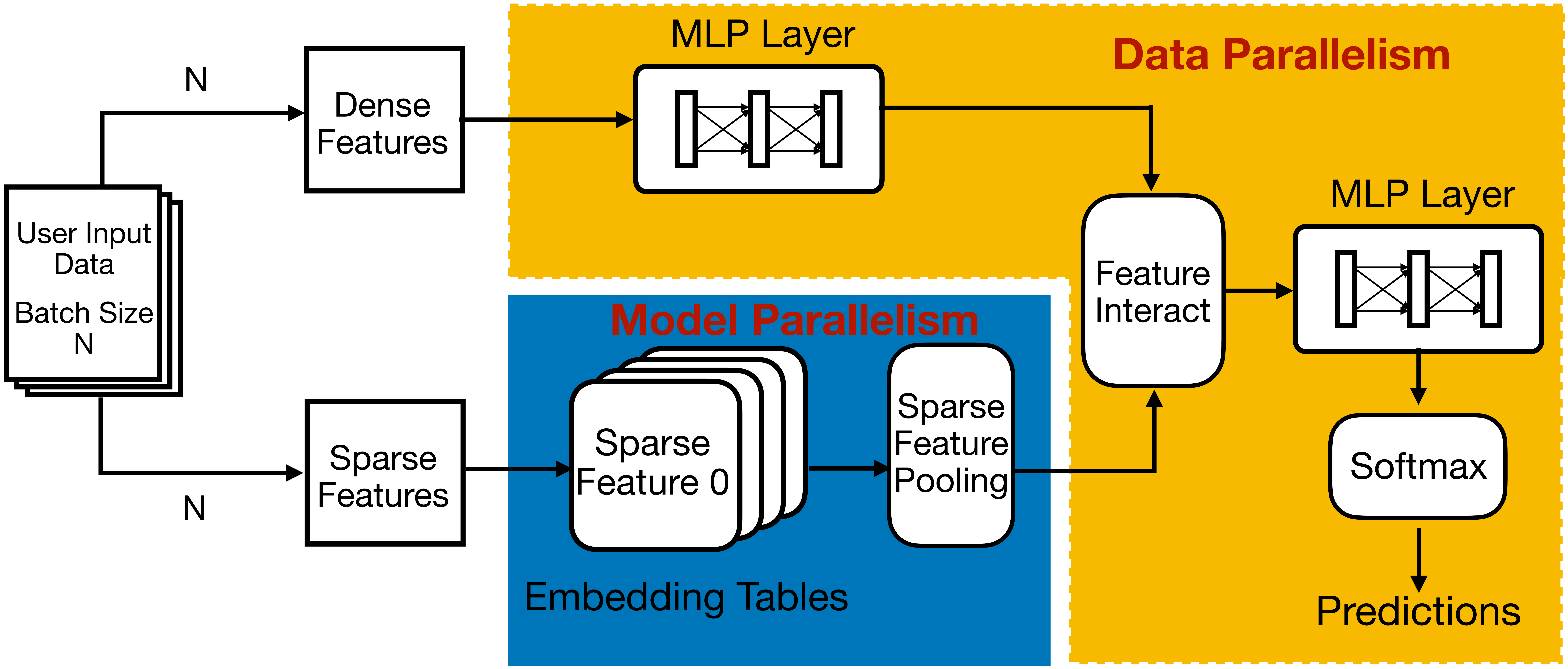

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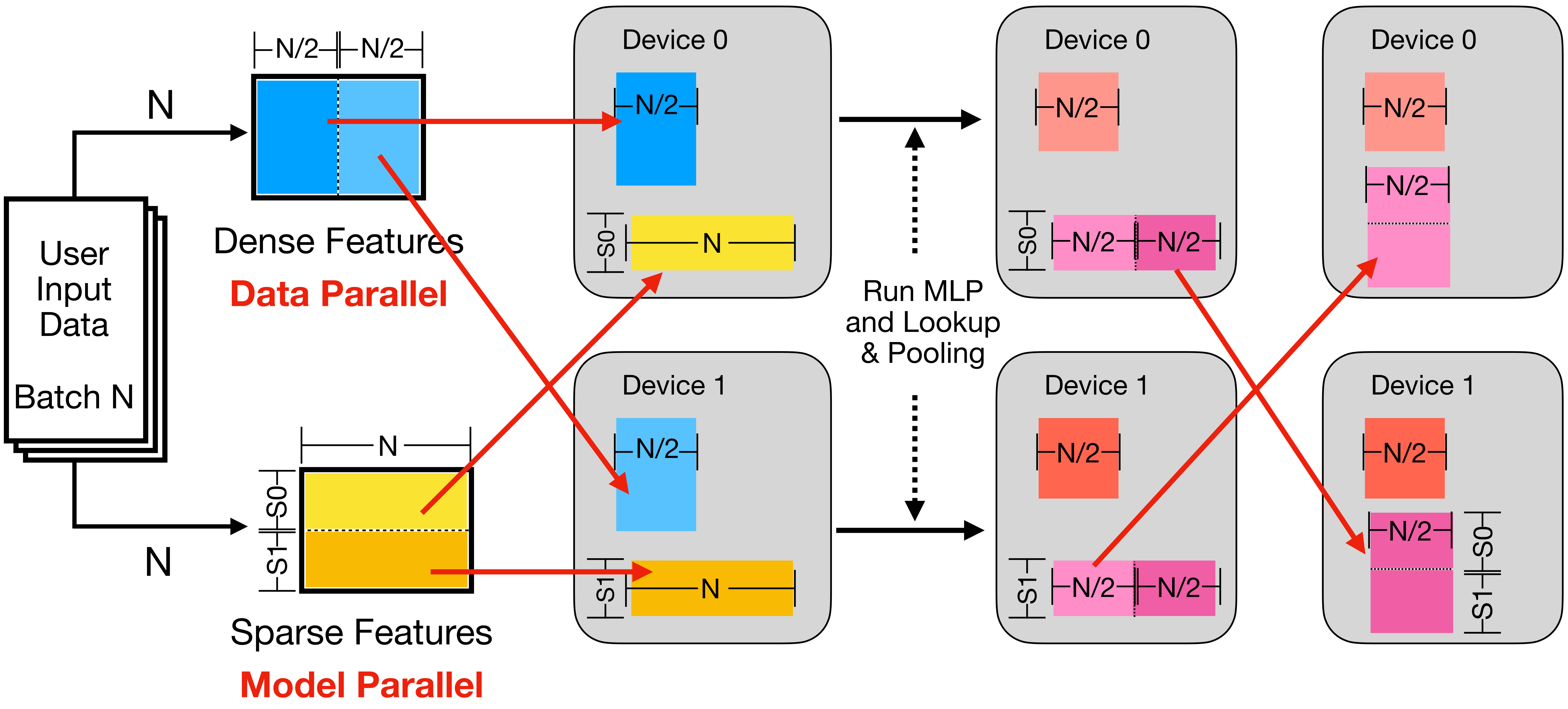

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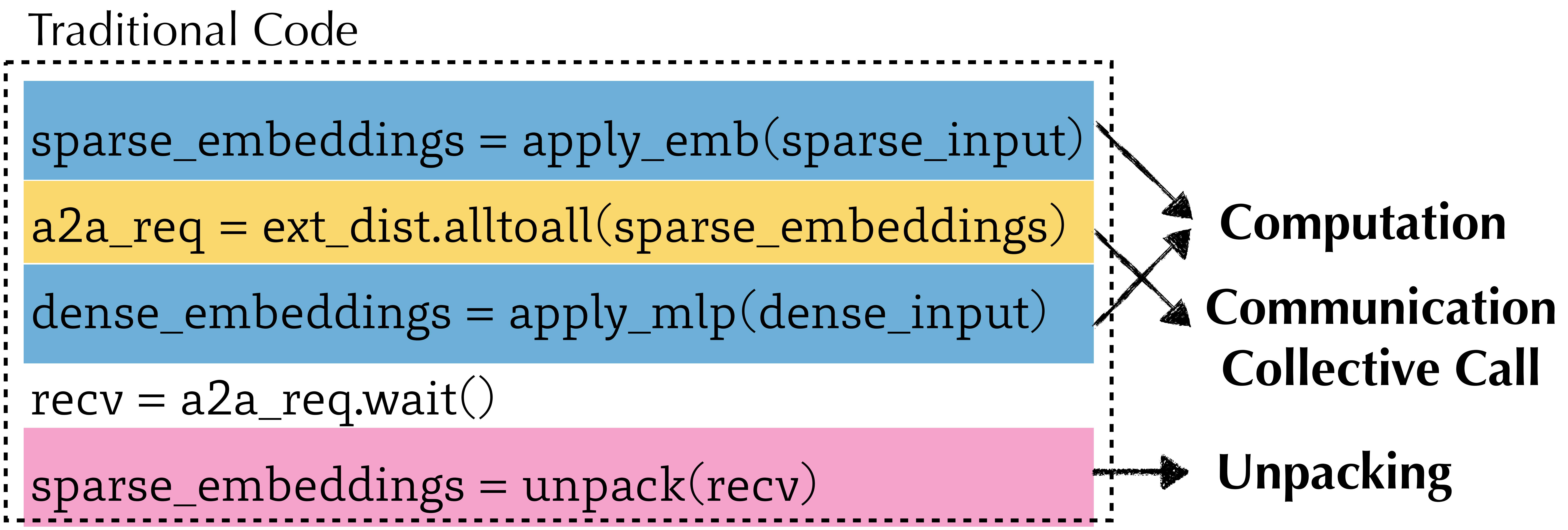

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PGAS One-sided Small Asynchronous Messages

Communication Schemes:

- **Small**
- **One-sided Async**
- **Inside GPU kernels**

Make Overhead Low

What is BENEFIT?

- **Remove Send Aggregation & Receive Unpacking, Writing to Targeting Memory Location Directly**

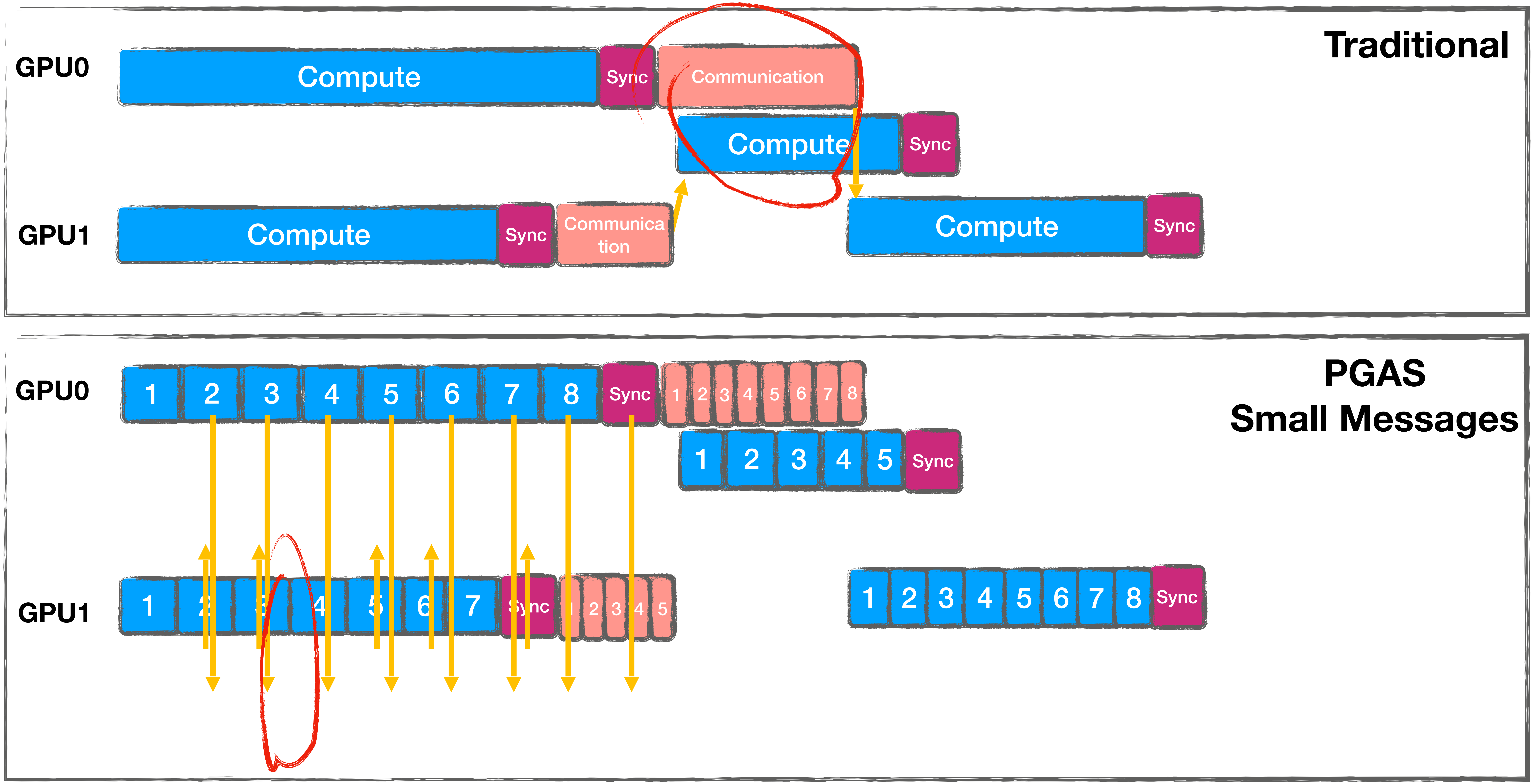
PGAS One-sided Small Asynchronous Messages

At the Time of Communication…..

slower/more overhead

faster/ less overhead

PGAS One-sided Small Asynchronous Messages



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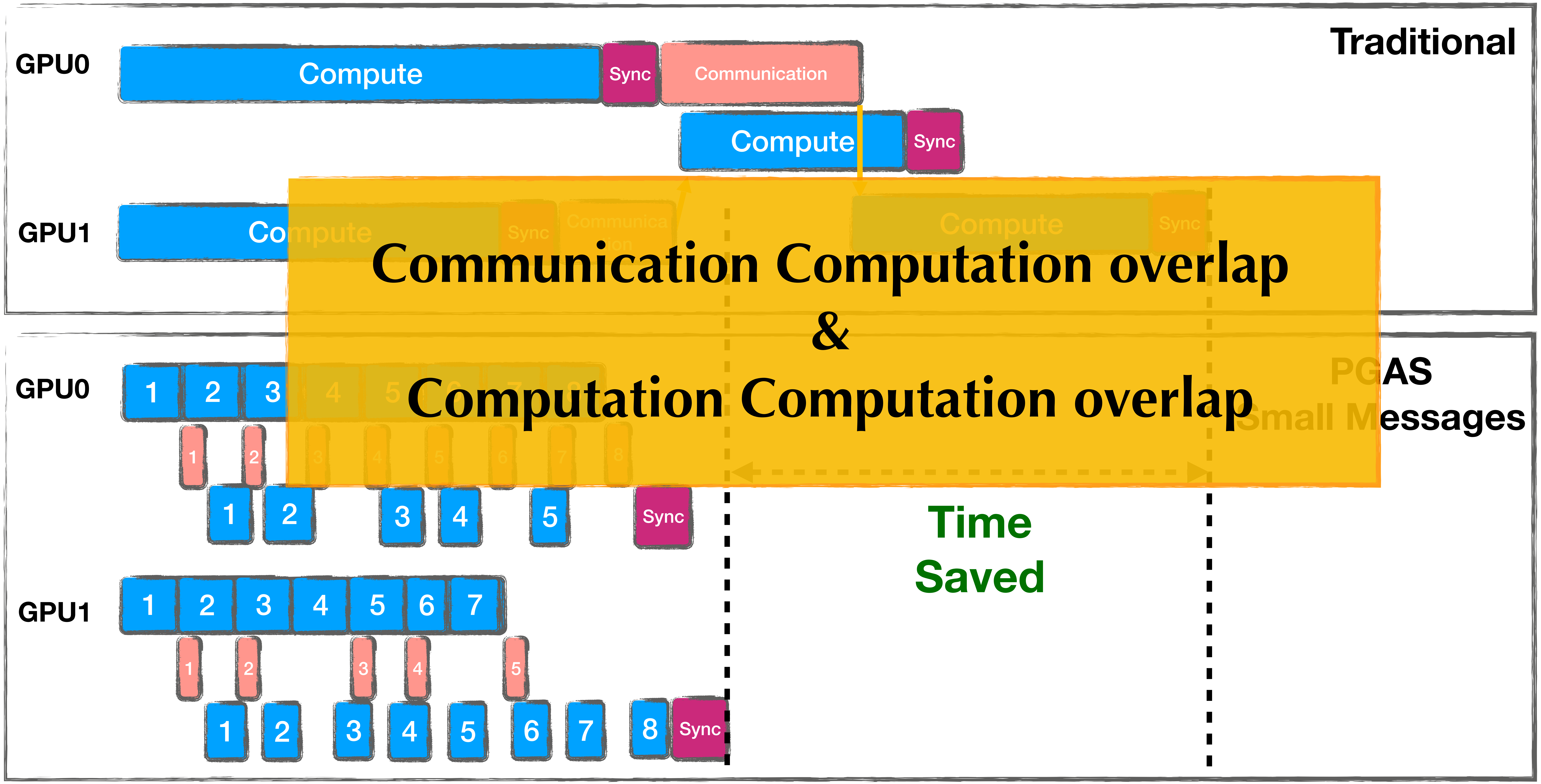
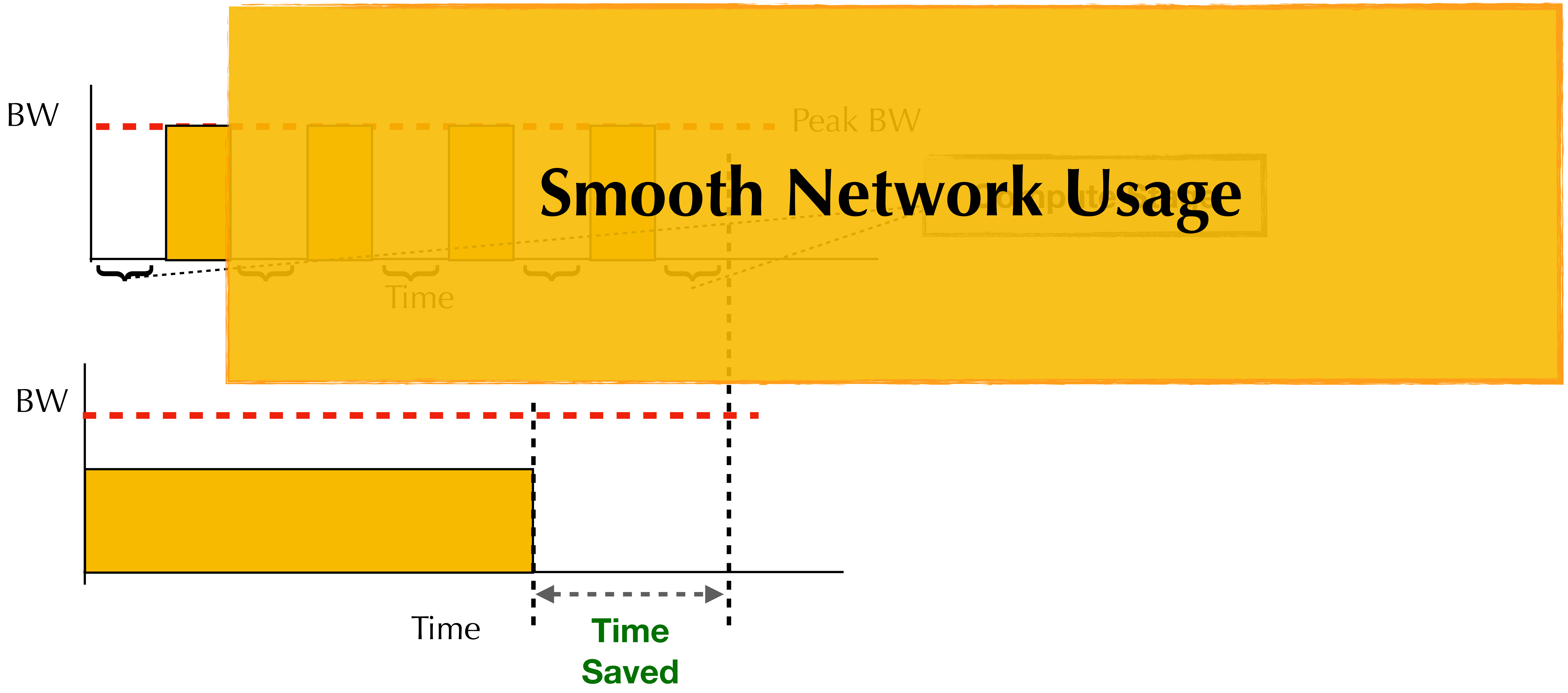


PGAS One-sided Small Asynchronous Messages

Smooth out the network usage by spreading communication along execution time.



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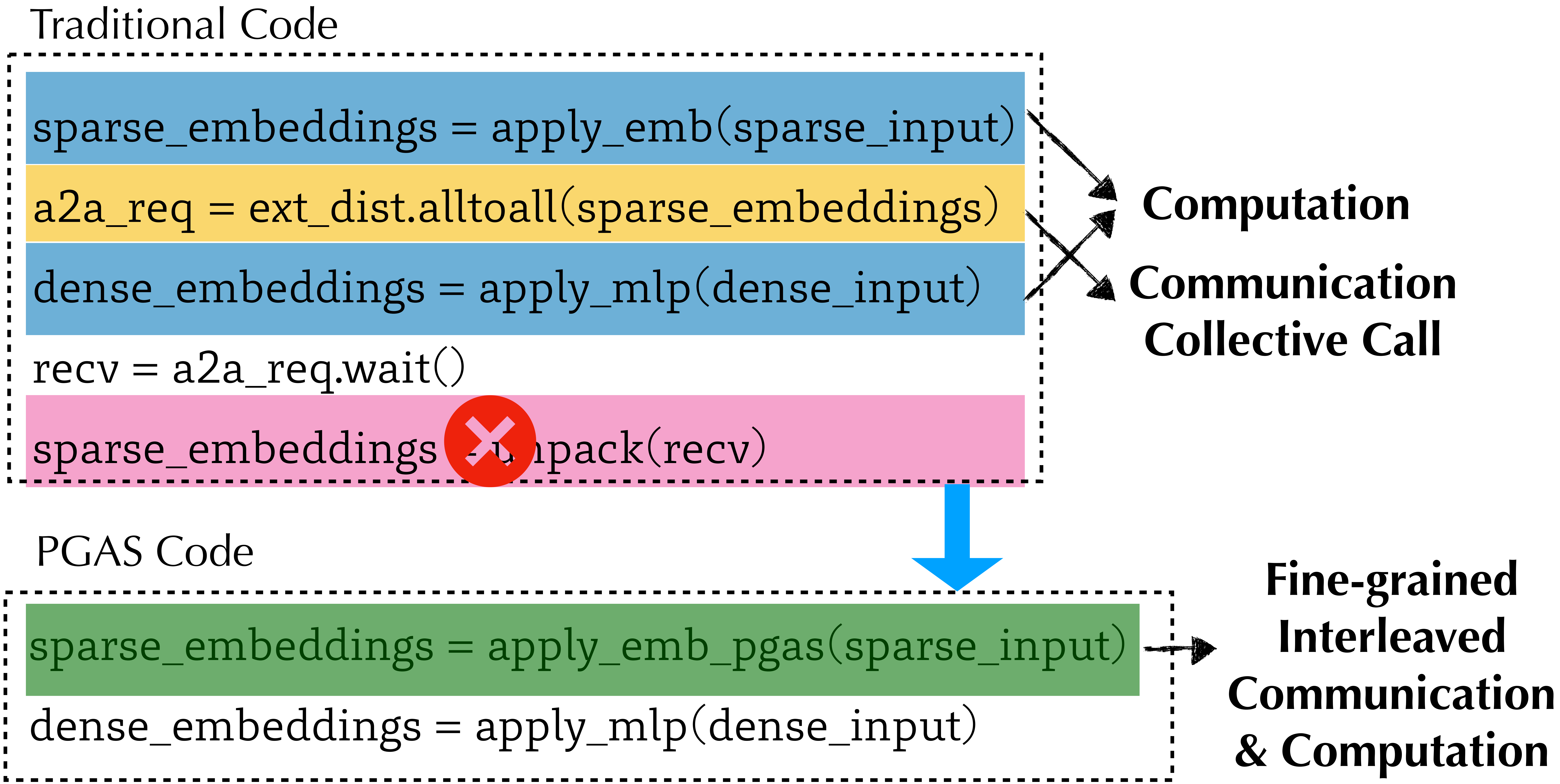

PGAS One-sided Small Asynchronous Messages

```
template<typename scalar_t, typename AccuFunc, typename ShardFunc>
__global__ void PGAS_EMB_forward_kernel(
    const PackedTensorAccessor64<scalar_t, 2,
                                 RestrictPtrTraits> weights,
    const PackedTensorAccessor64<int64_t, 1,
                                 RestrictPtrTraits> offsets,
    const PackedTensorAccessor64<int64_t, 1,
                       .....     RestrictPtrTraits> indices,
   ShardFunc GetEmbOwnerId) {
   // Compute the embedding lookup and pooling given sparse
   // input offsets and indices and store the embedding result
   // in Vec4T sum
   ......
   auto [pe, output_idx] = GetEmbOwnerId(
       threadIdx.y, blockIdx.x, BatchSize, n_gpus);
   if(pe == gpu_id) {
       // Writing the embeddings locally
       sum.store(outputs[output_idx]);
   }
   else {
       // PGAS writing the embeddings to remote GPU memory
       sum.store(outputs[output_idx], pe);
   }
}
```

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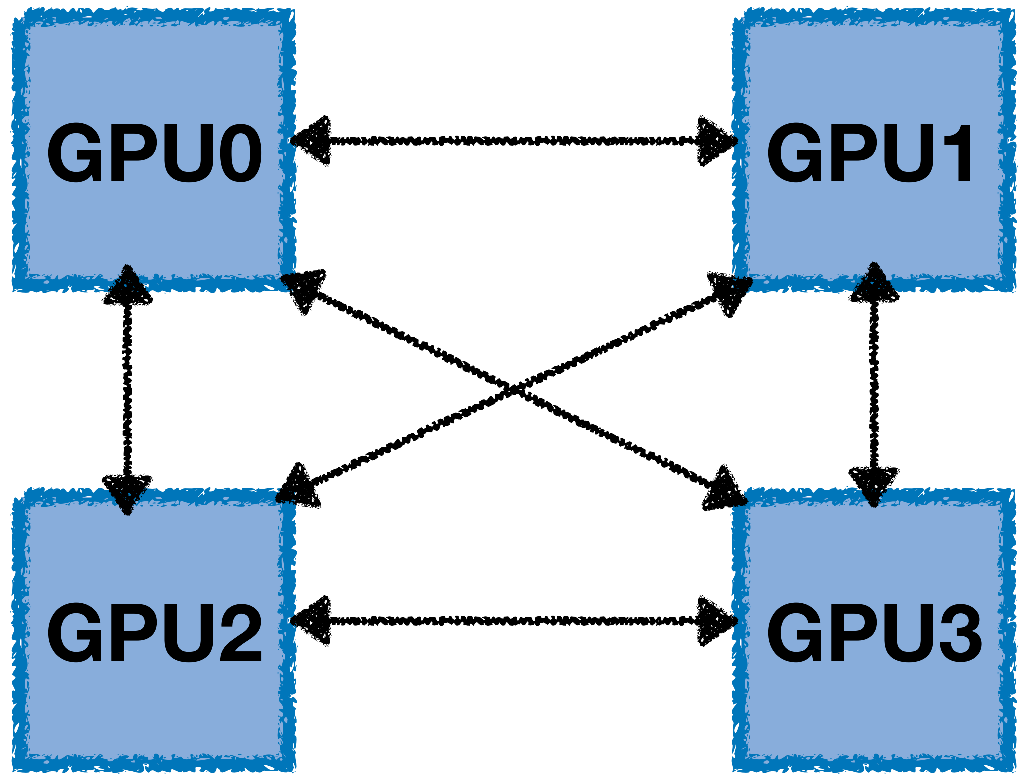
PGAS DLMR Case Studies

Case Studies:

- Computation vs (Communication+Unpack): 1/1
- Baseline: Pytorch+NCCL
- Our approach: PGAS fused implementation

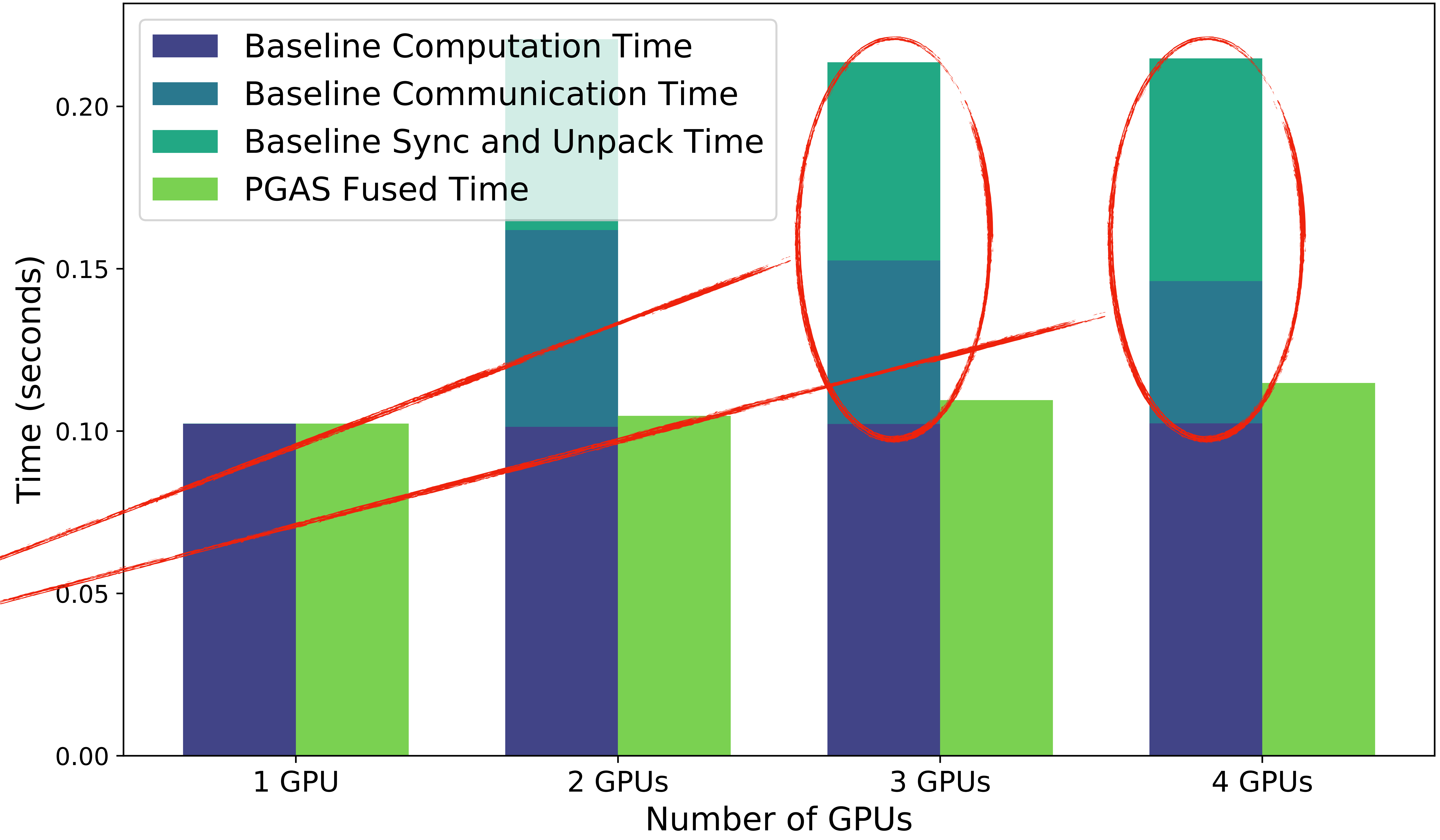
System: NVLink System



PGAS DLMR Weak Scaling

Weak Scaling

- Best overall: **PGAS** implementation
- **PGAS** reduces sync cost and eliminate the unpack operation
- **PGAS** is able to completely hide the communication with the computation



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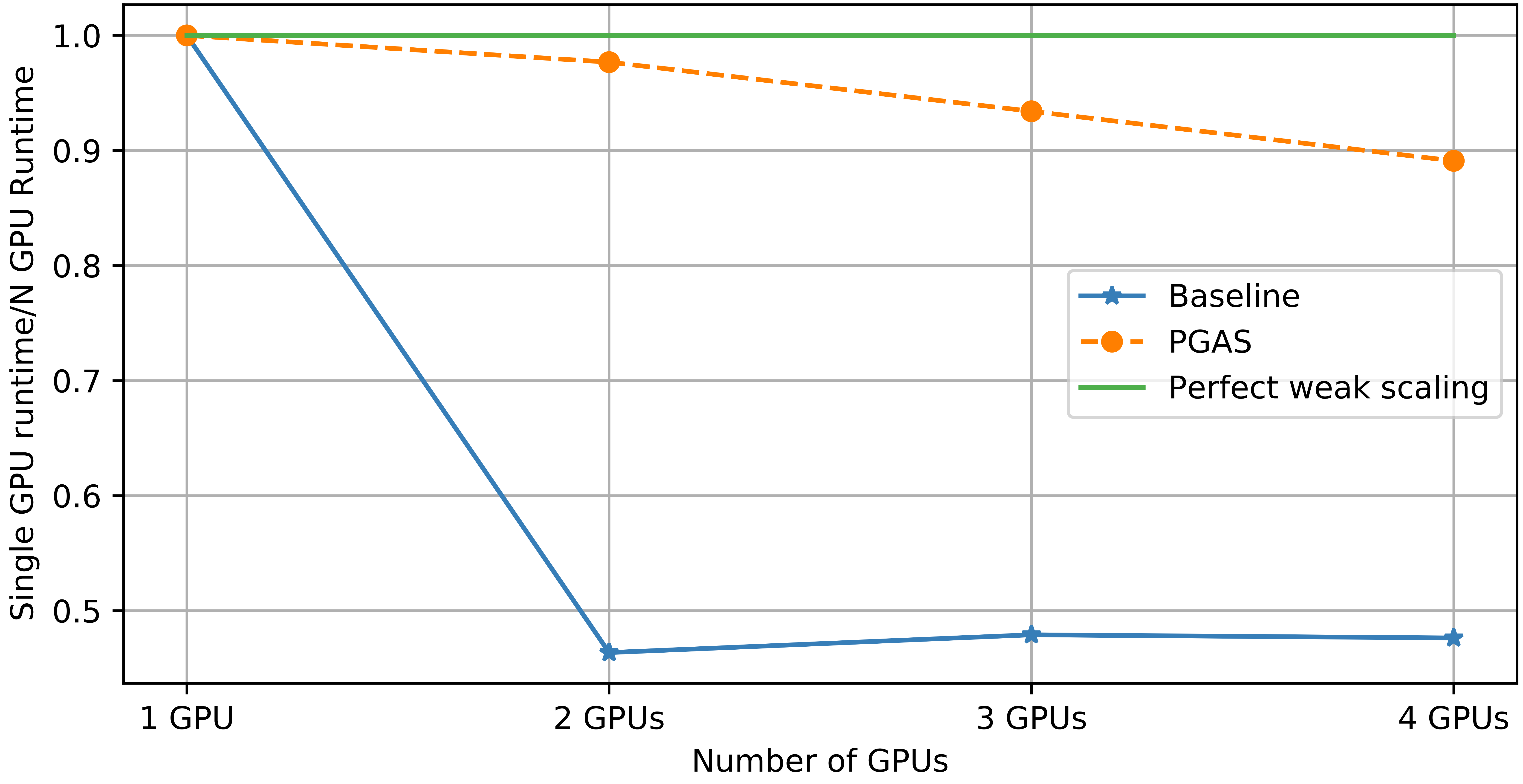

PGAS DLMR Strong Scaling

Strong Scaling

- Best overall: **PGAS** implementation
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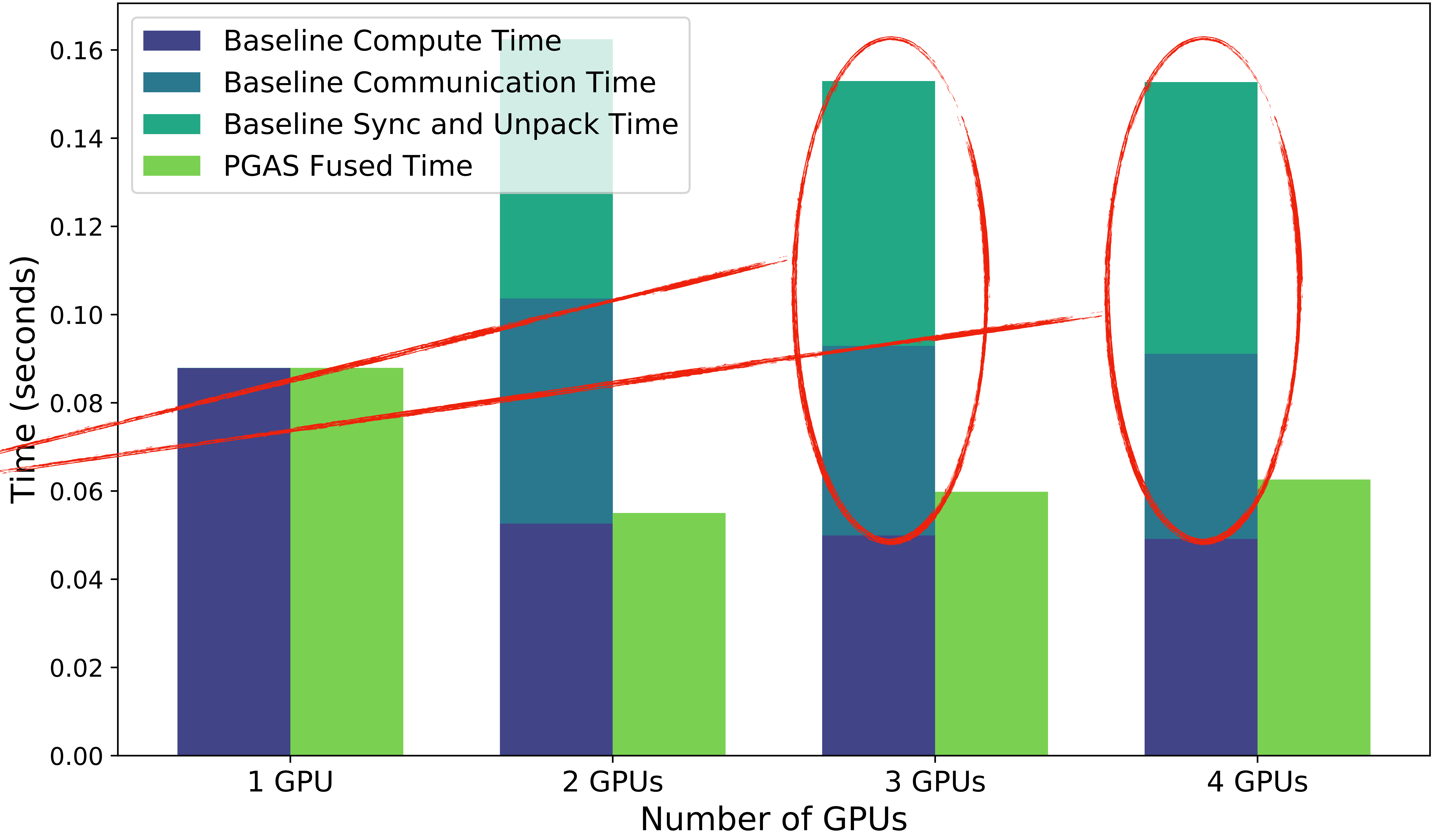
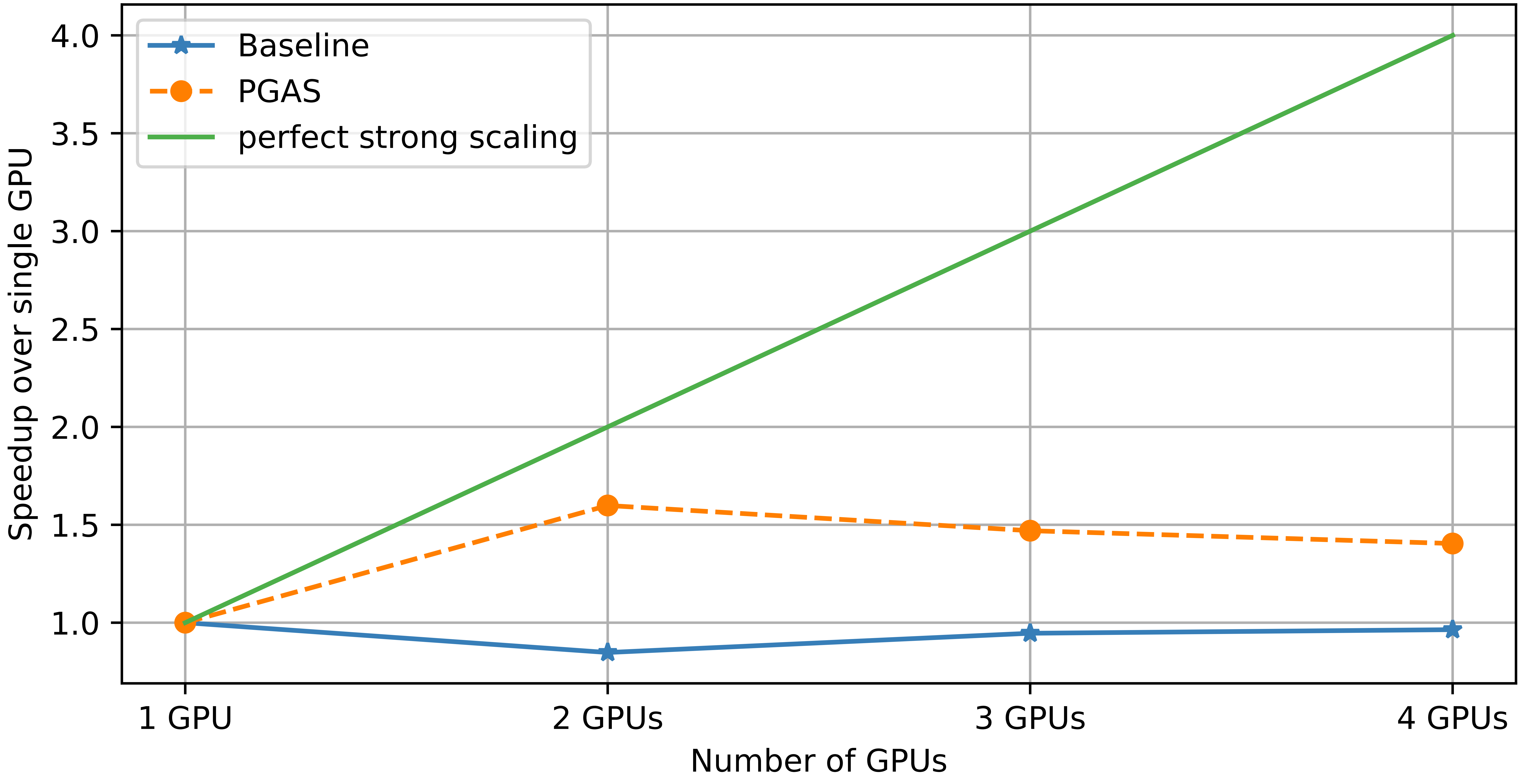
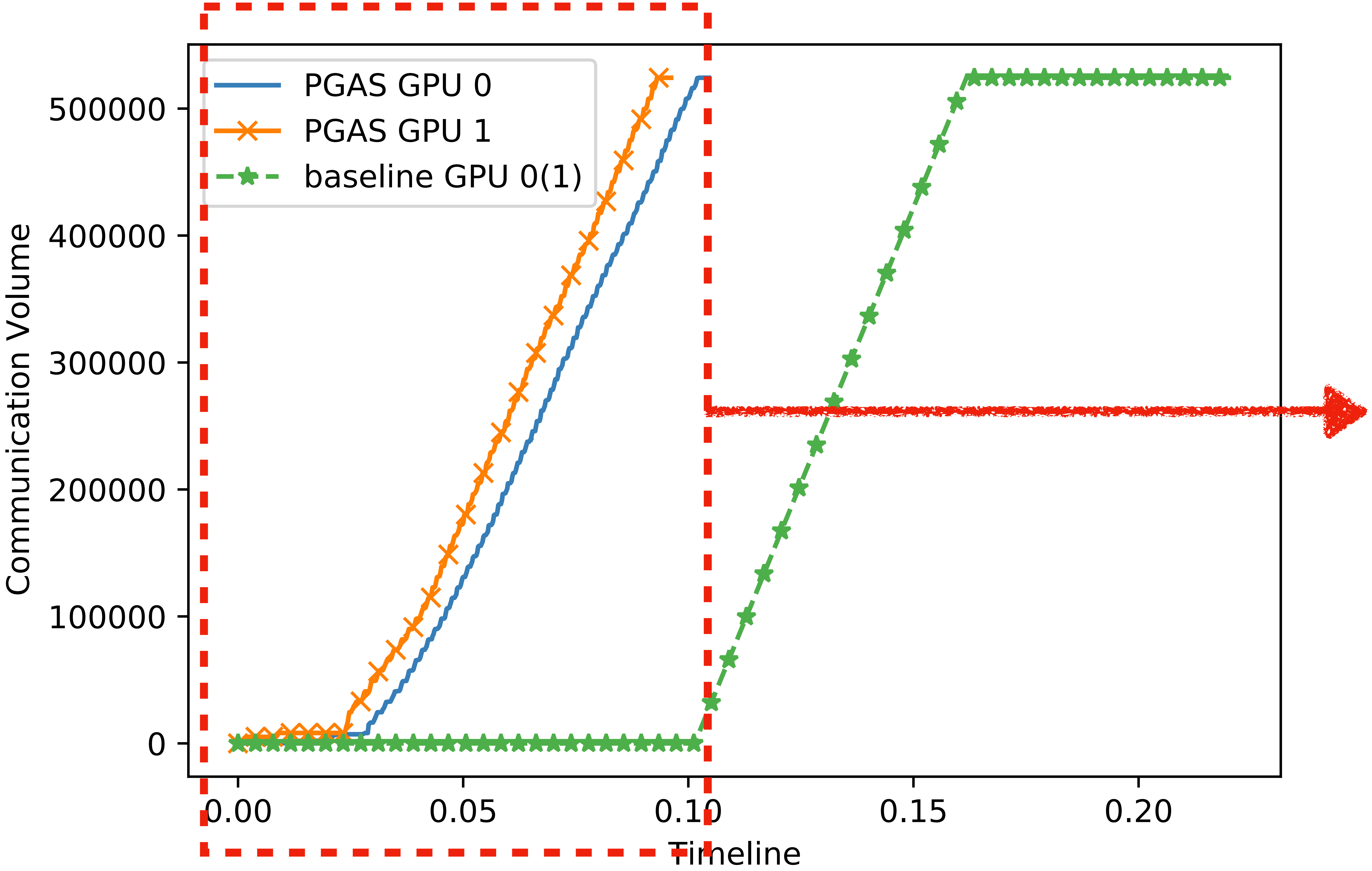


PGAS DLMR Strong Scaling

Strong Scaling



PGAS DLMR Communication Profile



Well-distributed over computation

Conclusion

PGAS one-sided async small messages performance benefits:

1. A GPU-centric control path reduces communication latency (issue inside kernel)

2. Fine-grained computation and communication overlap

3. Smooth network usage

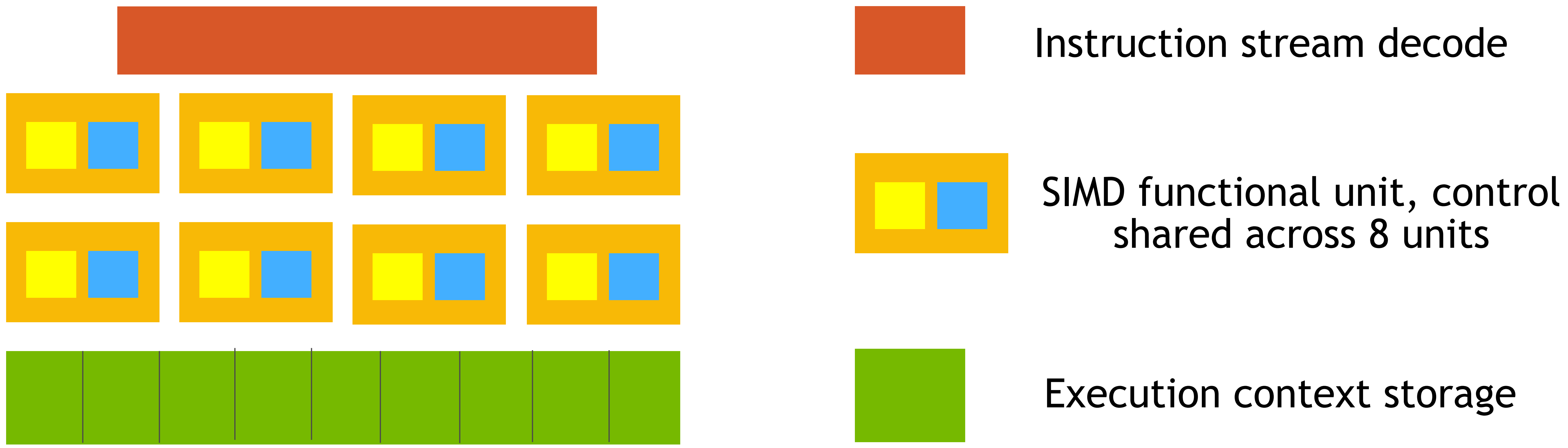
4. Eliminating unpack steps

Thank You

PGAS One-sided Small Asynchronous Messages

Muti-threading requires enough work to hide latency, Do we have enough work?

- Communication size is small thus easy to hide
- GPU uses same philosophy to hide stall



PGAS One-sided Small Asynchronous Messages

When to use PGAS one-sided small asynchronous messages?

- Computation vs Communication ratio
  - > 1:1 ratio ✔ 1:2 ratio ✔ 1: 100 ratio ✘
- Hardware used
  - NVLink ✔ IB band ✔ Ethernet ✘
- Memory Access Pattern, TLB size, etc.

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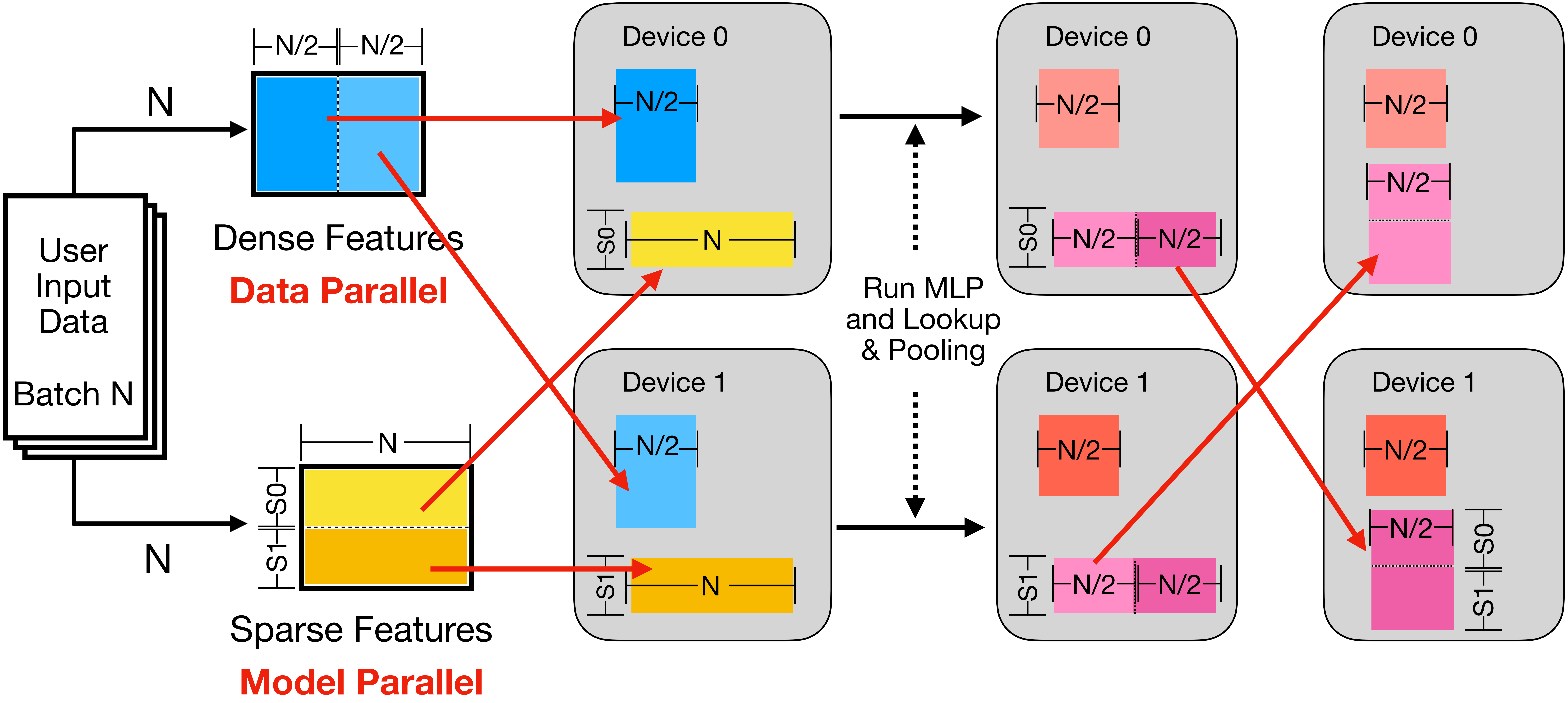

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